

Unit 5, Lesson 1: Determining Solutions to Inequalities

Benchmark

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning Targets

- ☐ I can determine which value from a set of values makes an inequality true.
- ☐ I can determine if a value on a number line represents a solution set.
- ☐ I can solve inequalities and represent solutions on a number line.

5.1.1 Warm-Up: Careers – Forester

Foresters have a wide range of duties, depending on who their employer is. Some primary duties of foresters are drawing up plans to regenerate forested lands, monitoring the progress of those lands, and supervising tree harvests. Another duty of a forester is devising plans to keep forests free from disease, harmful insects, and damaging wildfires.

1. Why do you think being an effective communicator and a team player is essential to the job?
2. Describe a time when you had to be an effective communicator and team player.

5.1.2 Exploration: Is it a Solution?

A solution to an inequality is any value that, when substituted for the variable, makes the inequality statement true.

A list of possible solutions for the inequalities $2x - 5 > 13$ and $2x - 5 < 13$ is shown.

Possible Solution Values			
9.01	−4000	11.47	0
$115\frac{3}{4}$	−9	9	−6.25
8.375	10	38	$4\frac{1}{6}$

1. Determine whether each of the possible values is a solution to the given inequalities.

a. Record the solutions in the table.

$2x - 5 > 13$	$2x - 5 < 13$	Not a Solution to Either

b. Explain the strategy you used to identify solutions.



An inequality has infinite solutions and can be represented on a number line.

2. Discuss with your partner why an inequality has infinitely many solutions.
3. Predict the solution set for $2x - 5 > 13$.
4. Predict the solution set for $2x - 5 < 13$.
5. There are no values for x that satisfy both inequalities. Discuss with your partner why that is the case. Summarize your discussion.
6. Neither inequality had a solution of 9. How could the inequalities $2x - 5 > 13$ and $2x - 5 < 13$ be rewritten so that 9 would be part of the solution set?

5.1.3 Guided Practice: Solving Two-Step Inequalities

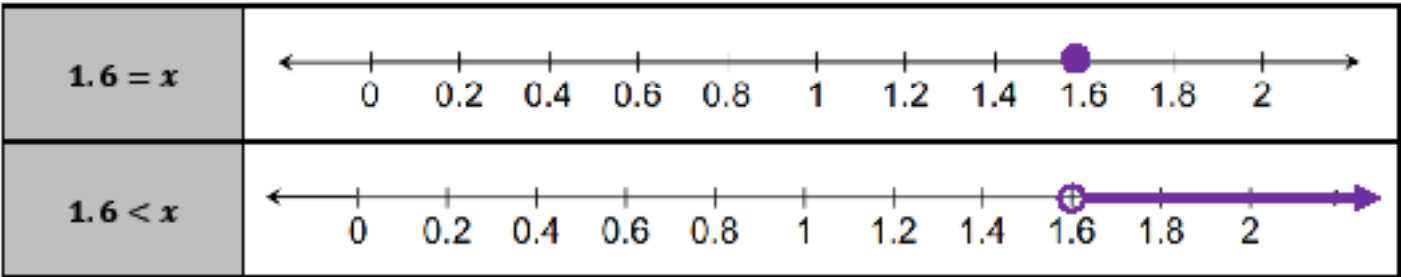
Inequalities can be solved using similar strategies that are used for solving equations.

1. The work for solving the equation $17 = 5x + 9$ and the work for solving the inequality $17 < 5x + 9$ is shown.

Equation	Justification	Inequality
$17 = 5x + 9$		$17 < 5x + 9$
$-9 \quad -9$		$-9 \quad -9$
$\frac{8}{5} = \frac{5x}{5}$		$\frac{8}{5} < \frac{5x}{5}$
$1.6 = x$		$1.6 < x$

- a. Complete the table with the justification of each step used for solving the equation and the inequality.

The solution for the equation and the solution set for the inequality is shown.



- b. Discuss with your partner the similarities and differences between solving and graphing the solutions for equations and inequalities. Summarize your discussion.

2. The work for solving the equation $13 = -5t - 7$ and the inequality $13 \geq -5t - 7$ is shown.

Equation	Justification	Inequality
$13 = -5t - 7$		$13 \geq -5t - 7$
$+7 \quad +7$		$+7 \quad +7$
$20 = -5t$		$20 \geq -5t$
$\frac{20}{-5} \quad \frac{-5}{-5}$		$\frac{20}{-5} \quad \frac{-5}{-5}$
$-4 = t$		$-4 \leq t$

- a. Complete the table with the justification of each step used for solving the equation and the inequality.
- b. How did the final step of solving the inequality $13 \geq -5t - 7$ differ from the final step of solving the inequality $17 < 5x + 9$?



When the variable term in an inequality is negative, the inequality can be rewritten by multiplying or dividing the inequality by -1 . In the resulting inequality, each term and the comparison symbol will change to its opposite.

3. Solve each inequality and graph the solution.

a. $2 - \frac{a}{6} \geq -4$

b. $13 \geq -5t - 7$

5.1.4 Your Turn!

1. Solve each inequality and graph the solution.

a. $56 \geq 8x + 21$



b. $-2.7 - 0.5x > 8.3$



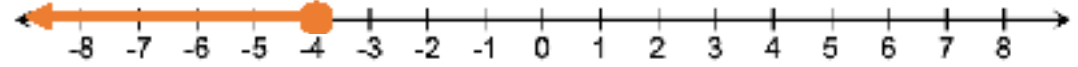
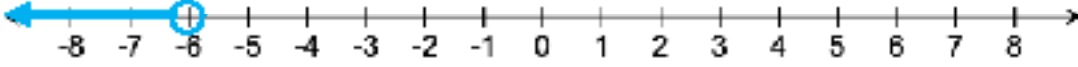
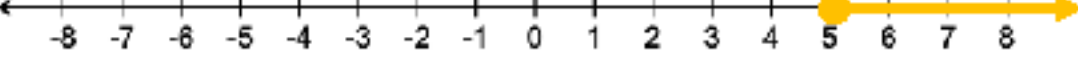
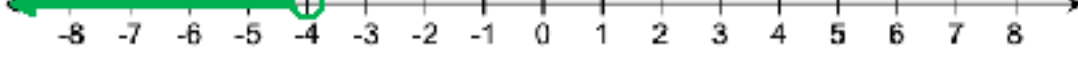
c. $6 + \frac{3}{4}k < -12$



5.1.5 Collaboration: Matching Inequalities

Work with your partner to match each of the given inequalities to their graph of possible solutions.

1. Record your matches in the table.

Graph	Inequality
	
	
	
	
	
	

2. What strategies did you use to match each inequality to its graph?

5.1.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	